DNP3 In-Network Fingerprint Normalization

Author Name(s)
Affiliation
Email address

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I. Introduction

Device fingerprinting is an essential step in cyber reconnaissance, letting adversaries reveal unique features of a target network and design stealthy attack strategies. These techniques are increasingly critical in industrial control systems (ICSs) such as power grids, where adversaries use IP-based control networks as entry points to inflict physical damage. Consequently, fingerprinting shifts the focus from identifying visited websites and user behavior to revealing device models and control operations that are critical to ICS attacks. In the 2015 attack that disrupted the Ukrainian power grid and the Stuxnet attack that disrupted Iranian nuclear facilities, it is widely believed that adversaries stayed in the target systems for at least six months to perform cyber reconnaissance [1].

Power grids carry their measurements and control commands over protocols such as DNP3 (Distributed Network Protocol 3) [2], which was deployed with no encryption by default [3]. Consequently, any observer on the communication path can read the exchange, and need not even decode the payload. Because a control-related attack, such as a false-data injection attack on state estimation, works only when the attacker knows the target device [4], the attacker first learns which devices communicate and how they behave. For this step, the response timing leaks enough on its own [5]: from it alone, an observer can identify a device, its type, and its model [6], [7].

To disrupt device fingerprinting, many studies present network traffic obfuscation [8], [9]. These defenses target the network-level features that fingerprinting relies on [10], in two directions. First, padding and splitting change the distribution of packet sizes [8]. Second, delaying packets and adding dummy ones change the inter-packet timing [9], [11]. Since manipulating traffic at the host adds runtime overhead, recent work offloads obfuscation onto programmable switches, which reshape communication at much higher line rates than CPUs [12]–[14].

Unfortunately, these methods do not transfer to device fingerprinting in ICS environments, for two reasons. First, the leaked features differ. General obfuscation targets flow-level size and timing patterns [12], whereas an ICS device fingerprint is carried by the response timing of the protocol exchange, which stays visible even when the payload is encrypted [5]. Second, ICS protocols leave little room to add bytes or fabricate traffic. Because DNP3 validates its payload, padding and injected packets break the exchange: the master reassembles each response and checks a cyclic redundancy check for every sixteen application bytes [2]. Consequently, a defense for this setting must hide timing that stays visible under encryption while touching no byte.

In this paper, we present SYSNAME, an in-network defense that meets both conditions. SYSNAMEnormalizes an outstation's fingerprintable timing in the data plane of a single programmable switch at the outstation edge, without changing the endpoints and without modifying a DNP3 byte. For a read, it holds the acknowledgment and the response in switch queues and releases them at fixed offsets from the request, so the master-visible timing becomes a single policy value. For a control command, it normalizes the timing on a second, independent scheduling lane, so the operation time also becomes a single value. Because SYSNAMEonly holds and forwards packets, it changes no byte. We implement it as one P4 program on a single switch pipeline and evaluate it on a physical SEL-751 relay.

We make the following contributions.

- **Normalizes ICS device-fingerprinting timing in the data plane.** We propose an in-network framework that removes the timing features an on-path observer uses to fingerprint a DNP3 outstation, using one switch at the outstation edge and changing no endpoint and no DNP3 byte. To the best of our knowledge, this is the first in-network defense that normalizes ICS device-fingerprinting features on real hardware.
- **Normalizes the cross-layer response time.** As a first case study, reproducing the read-side feature that Formby et al. use [5], we hold the acknowledgment and the response and release them at fixed offsets from the request, driving the master-visible timing to a single policy value.
- **Normalizes the control-command timing on an independent lane.** As a second case study, covering the physical-operation feature of the same attack [5], we hold the control response, pin the master-visible acknowledgment and echo to the request, and show why the read and control paths need two independent scheduling lanes.
- **Evaluates on a physical relay.** We evaluate SYSNAMEon a physical SEL-751 with a reproducible capture-to-claim analysis, an explicit safety result, and a stated evidence boundary.

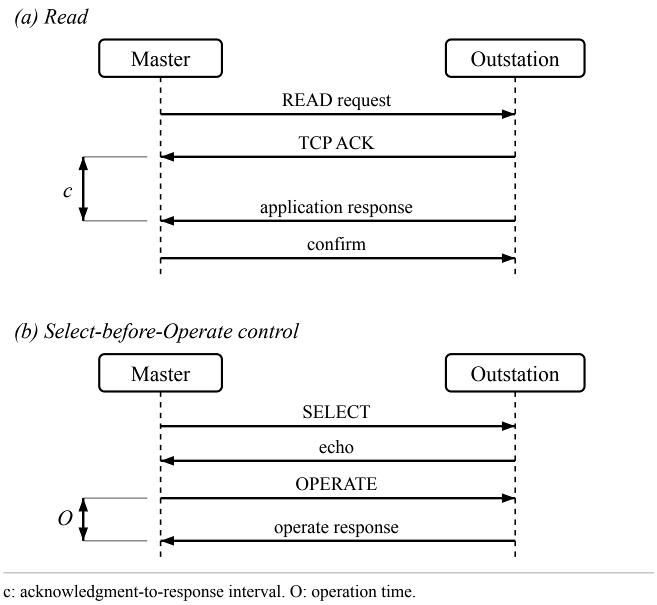

Fig. 1. DNP3 transactions on the SEL-751. A read is a request, a transport-layer acknowledgment, and an application-layer response; a Select-before-Operate control adds an OPERATE and its response. The adversary times the acknowledgment-to-response interval of the read and the operate-to-response interval of the control.

II. Background

DNP3 polling and control. DNP3 carries SCADA reads and controls over TCP between a master and an outstation [2]. The master periodically polls the outstation with a read request, and the outstation returns the requested measurements in a response. A control uses Select-before-Operate: the master sends a SELECT that names an output point, the outstation echoes it, and the master then sends an OPERATE that actuates the point, which the outstation confirms. Figure 1 shows both exchanges.

Cross-layer timing on a separate-acknowledgment device. Our outstation is a physical SEL-751 feeder-protection relay. Because TCP acknowledges a request at the transport layer before the application response is ready, each transaction exposes two events to an on-path observer: the acknowledgment and, later, the response. Consequently, a read exposes the interval between its acknowledgment and its response, and a control OPERATE exposes the interval between its request and its operate response. These two intervals reflect the outstation's internal processing rather than the network path, and they are the timing features the adversary uses (Section III).

Programmable data planes. Modern switches expose a match-action pipeline, programmable in P4, that parses headers, keeps per-packet state, and holds packets in output queues at line rate. Since this pipeline can delay and release a packet without a host in the loop, it is where SysNameruns, changing no endpoint and no DNP3 byte.

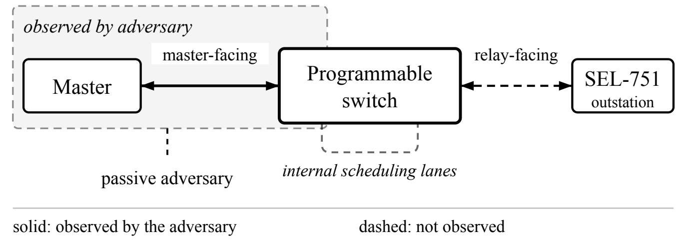

Fig. 2. Observation model. The master reaches the SEL-751 outstation through one programmable switch at the outstation edge. The passive adversary observes only the master-facing link; the relay-facing link and the switch-internal scheduling lanes are not observed.

III. Threat Model and Objectives

Assumptions on the adversary. We adopt the threat model of Formby et al. [5]. Figure 2 shows the setup. The adversary is passive: it observes the master-facing link, records TCP and IP headers, segment sizes, and arrival times, and neither modifies nor injects a packet. Because legacy DNP3 usually runs without transport security across shared substation networks [3], we assume the traffic is unencrypted. Consequently, the adversary reads the exchange directly, and its advantage comes from observing it, not from a secret.

Reconnaissance target. The adversary fingerprints the outstation to prepare a later attack. Because a control-related attack, such as a false-data injection attack, works only when the attacker knows the target device [4], this reconnaissance is the enabling step. The fingerprint survives without decoding the payload, because it is carried by the timing of the exchange. Following Formby et al. [5], two timing features are in scope: the interval between the transport-layer acknowledgment and the application-layer response on a read, i.e., the cross-layer response time (CLRT); and the interval between a control request and its response, i.e., the operation time. The first reflects the outstation's internal processing, and the second reveals the physical device behind the control command.

Objectives. Our objective is to remove these two features from the master-facing observation. We state it as three properties that anchor the design and the evaluation.

- **RO1 (read timing).** The master-visible CLRT is a fixed policy value, independent of the transaction, so it no longer reflects the outstation's native processing.
- **RO2 (control timing).** The master-visible operation time is a fixed policy value, independent of the switch-internal release delay.
- **RO3 (safety).** The defense actuates no physical output.

For active and relay-facing adversaries. An adversary that manipulates traffic, breaks a protected channel, or observes the relay-facing link is out of scope. Passive observation of the master-facing link is the setting that prior traffic-obfuscation work targets and the one an unencrypted deployment exposes; we leave the others to future work.

For packet-size fingerprinting. We do not obfuscate the outstation's response length. Our scope is the two timing

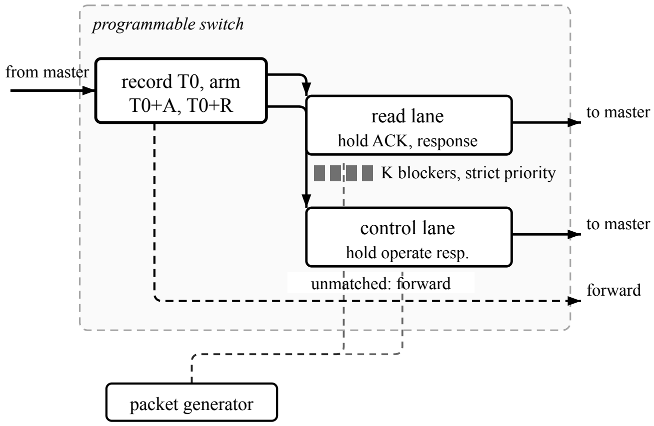

Fig. 3. Design of SysName. Ingress anchors every deadline to the request arrival T_0; two independent scheduling lanes hold the read and the control answers; reservoirs of blocker packets, drained by a strict-priority scheduler, realize the deadlines without a timer.

features above, and the response length is not a target of this defense.

IV. Design

In Figure 3, we present the design of SysName. A request enters the switch from the master. The program records the request's arrival time and forwards the request to the relay. When the relay answers, the program holds the answer in a switch queue and releases it at a fixed offset from the recorded request time, so the master sees a timing that the policy sets rather than one the outstation produces. Because the read and the control answers must not interfere, each is held on its own scheduling lane. The rest of this section presents the four mechanisms this requires.

Anchoring deadlines to the request. A held packet must leave at a fixed offset from the request, but the switch has no general-purpose timer. On the request, the program records an ingress timestamp T_0 and arms two deadlines from the policy offsets A and R,

$$t_{\mathrm{ack}} = T_0 + A, \qquad t_{\mathrm{resp}} = T_0 + R, \tag{1}$$

at which it releases the acknowledgment and the response to the master. Consequently, the interval the adversary measures is

$$\Delta = t_{\mathrm{resp}} - t_{\mathrm{ack}} = R - A, \tag{2}$$

a policy value, in place of the native interval c that the outstation's internal processing produces.

Let a and r denote the arrival times, relative to T_0, of the relay's acknowledgment and response. Because the switch can delay a packet but cannot emit one before it exists, the offsets are admissible only if

$$A \geq a_{\max}, \qquad R \geq r_{\max}, \tag{3}$$

where $a_{\max}$ and $r_{\max}$ bound the native arrivals over the traffic to be protected. Since Δ is then constant for every transaction, the observable no longer varies with c.

Security Argument. The anchor matters as much as the offsets. Because the relay's own acknowledgment arrives at $T_0 + a$, a deadline of the form $T_0 + a + D$ would make the observable depend on a and carry the device's own timing into it. Anchoring to T_0, which the adversary already observes, removes that dependence and makes (2) independent of the outstation. We check this against a mutant design that arms the deadline at the relay's acknowledgment (Section V).

Reservoirs as data-plane timers. To realize a deadline from packet scheduling alone, a held packet waits in its queue behind a reservoir of small blocker packets. Since a strict-priority scheduler drains the blockers first, the held packet cannot leave until its reservoir empties. If τ is the time to drain one blocker, a reservoir of K blockers holds a packet for $K\tau$, so the control plane realizes a deadline D with

$$K = \lceil D/\tau \rceil, \tag{4}$$

and sizes the reservoirs of (1) with $D = A$ and $D = R$. The switch's on-chip packet generator fills the reservoirs, and it never generates traffic toward the master or the relay. A tag byte distinguishes the generator profiles, such that a read and a control trigger different reservoirs without ambiguity.

Two independent scheduling lanes. The read answer and the control answer are held at the same time, and a single lane would serialize them: whichever holds longer would delay the other and reshape the timing the policy fixed. Consequently, SysNamegives each treatment its own internal recirculation lane and its own queues, so the two holds proceed independently.

Normalizing the control command. A control uses Select-before-Operate, so the adversary times the interval between the OPERATE and its response. SysNameadmits the OPERATE once, holds it, and releases it to the relay at an internal delay J, while the acknowledgment and the echo the master sees are placed at fixed offsets from the OPERATE's arrival T_0, as in (1). The observable operation time is therefore

$$O = (T_0 + R) - (T_0 + A) = R - A, \tag{5}$$

in which J does not appear.

Security Argument. By (5), the observable operation time does not vary with J, because the master-visible events are anchored to the OPERATE's arrival rather than to the relay's answer. Consequently, an adversary that measures the operation time learns the policy, not the physical device behind the command. Because J is absent from (5), the adversary also cannot subtract a known offset to recover the device's own timing.

Failing open. A defense on the control path must not become an availability risk. If a transaction does not match an armed treatment, SysNameforwards it unchanged rather than dropping it. Consequently, an unexpected exchange loses its protection but not its delivery, and the switch actuates no output of its own.

Byte preservation. SysNameonly holds and forwards packets. Because it neither pads a response, injects a packet into the protected exchange, nor edits a DNP3 field, every byte the master reassembles is the byte the relay produced, and the payload's cyclic redundancy checks remain valid.

V. Evaluation

A. Effectiveness in RO1 (read timing)

B. Effectiveness in RO2 (control timing)

C. Safety in RO3

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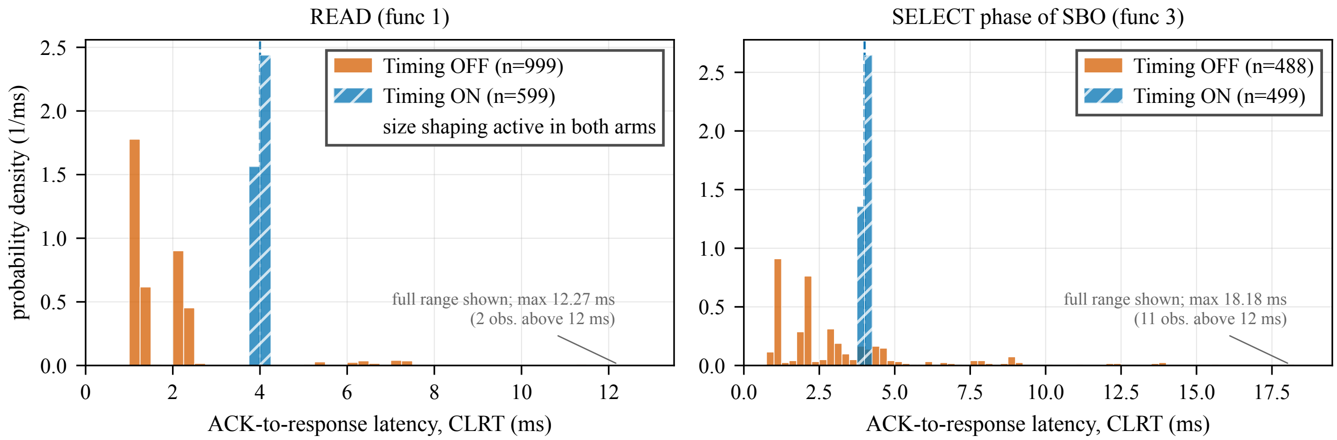

Fig. 4. Cross-layer response time (CLRT), the interval from the outstation's TCP acknowledgment to its DNP3 application response, with the timing mechanism disabled (Timing OFF) and enabled (Timing ON), measured master-facing against the physical SEL-751. Left: READ (function 1). Right: the SELECT phase of select-before-operate (function 3); these are SELECT observations, not complete SBO transactions. With the timing mode off, CLRT is spread over roughly 1 to 18 ms and differs between the two transaction types; with it on, both collapse onto a 4.001 ms policy value (dashed line) with a standard deviation of about 0.02 ms. Each panel's horizontal axis runs to its largest observation, so the full tail is visible. Both arms ran the same unified switch binary with the size-shaping datapath active; the Timing OFF arm is therefore not an unmodified device baseline, and the comparison isolates the timing-mode change.

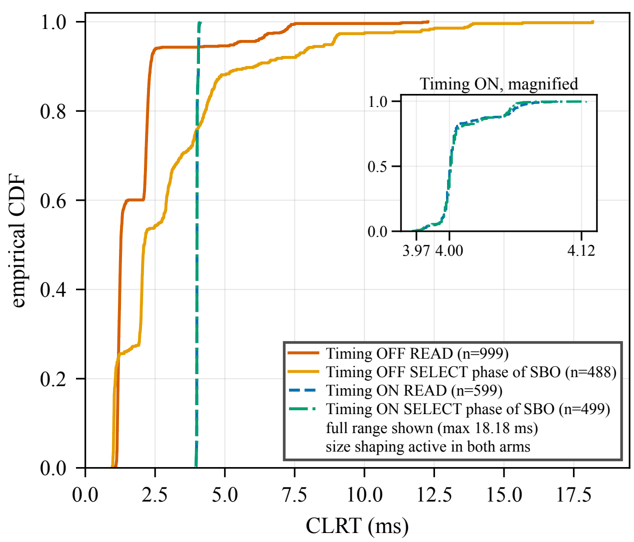

Fig. 5. Empirical CDF of CLRT for READ and for the SELECT phase of SBO with the timing mechanism disabled (solid) and enabled (dashed). With the timing mode off the two transaction types separate clearly and the SELECT distribution carries a tail to 18.18 ms, shown in full; with it on, both collapse onto 4.001 ms, and the inset resolves the two Timing ON curves, which coincide at full scale. Size shaping was active in both arms.

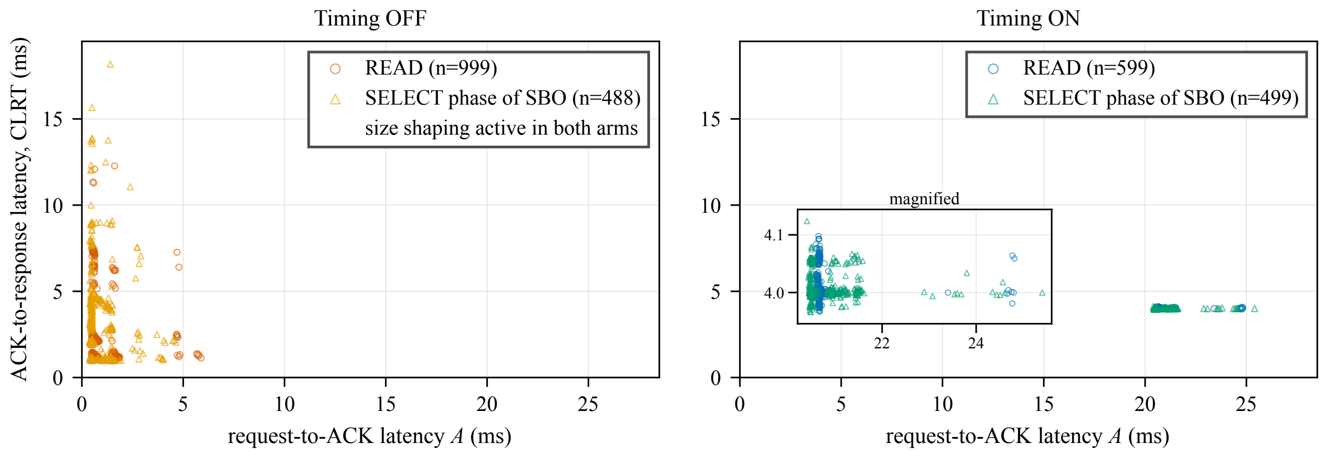

Fig. 6. Timing-feature overlap with the timing mechanism disabled and enabled. Each point is one transaction, placed by the two latencies a passive observer can measure master-facing: request-to-ACK on the horizontal axis and ACK-to-response (CLRT) on the vertical. Left, timing mode off: READ and the SELECT phase of SBO occupy visibly different regions. Right, timing mode on: both transaction types collapse onto a single point and are no longer separable by these features. This is a display of feature overlap between two transaction types on one relay, not a clustering result and not device identification; no unsupervised algorithm is fitted.

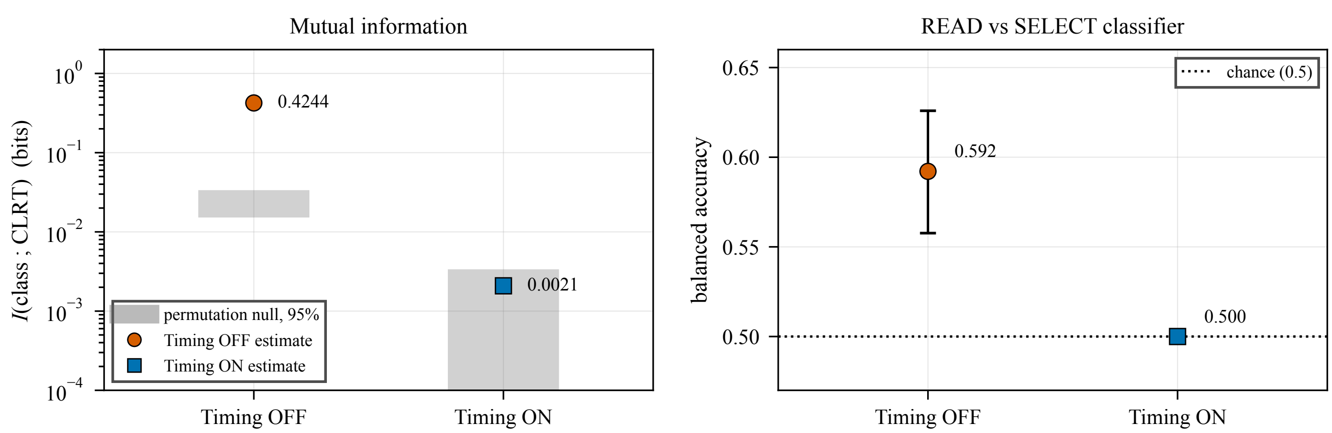

Fig. 7. Timing-only leakage with the timing mechanism disabled and enabled. Left: mutual information between transaction class and CLRT, with the shaded band giving the 95 % permutation null for each arm; with the timing mode off, CLRT carries about 0.42 bits about whether a transaction was a READ or the SELECT phase of an SBO, and with it on the estimate falls inside its own null band. Right: balanced accuracy of a classifier trained on Timing OFF CLRT to separate the two transaction types, applied unchanged to Timing ON traffic, falling from 0.592 to chance. Both panels describe transaction-class feature suppression on a single SEL-751 in a single capture session with a transaction-disjoint, not session-disjoint, split; neither is a device-identification result.

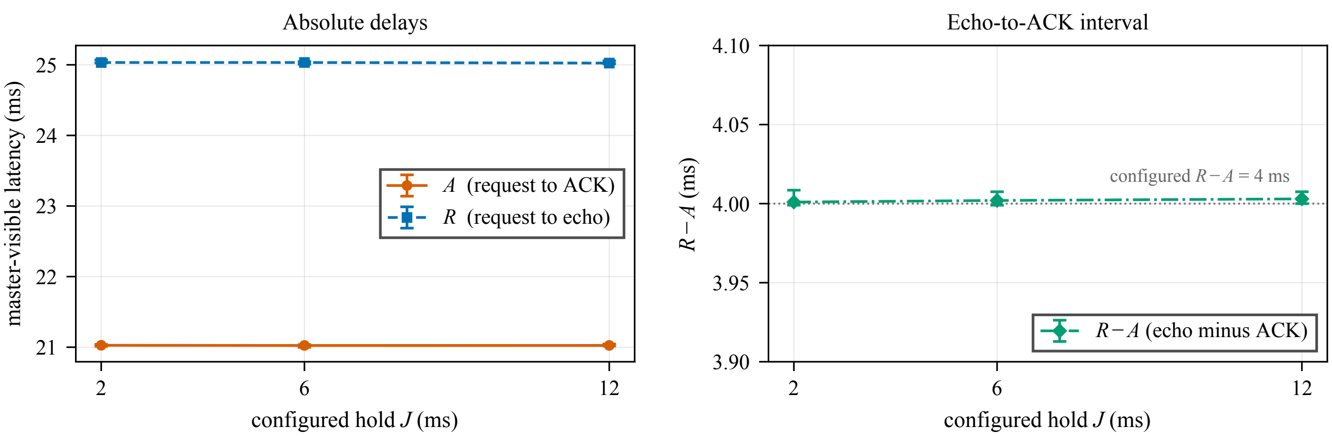

Fig. 8. OPERATE timing under three configured hold values. Left: the master-visible request-to-ACK delay A and request-to-echo delay R, both flat in J. Right: the echo-to-ACK interval $R - A$, which stays at about 4.00 ms across $J = 2$, 6 and 12 ms, so an observer who subtracts the two observable timestamps learns nothing about the configured hold. Markers are medians of 30 OPERATE transactions per condition; bars are bootstrap 95 % confidence intervals on the median. J is the configured codebook value; it was not observed on the relay-facing wire, and exactly-once relay delivery is not demonstrated.